Audit of a Fixed-Representative Reduction
and a Continuous Realization of All Pseudonorm Codes

Research note on Questions 1 and 2 of Cúth–Doležal–Doucha–Kurka

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Abstract

Cúth, Doležal, Doucha and Kurka asked whether every infinite-dimensional pseudonorm code can be converted continuously into a genuine norm code for the same Banach space, and whether one can at least obtain a $\mathbf{\Sigma}_2^0$ reduction to an admissibly topologized space of closed subspaces. This note audits a proposed fixed-subsequence argument and finds a genuine flaw: it tests only rational linear independence, whereas membership in $\mathcal{B}$ requires real linear independence after extension to real c_{00}. An explicit irrational-slope counterexample is given, followed by the corrected fixed-stratum theorem.

A literature search through the date above did not locate a published or arXiv resolution of either stated question. The main result of the note is a direct construction of continuous maps $\chi_n : \mathcal{P} \to C(Q)$, for one fixed compact metric space Q, satisfying

$$\left\| \sum_{n=1}^{N} a_n \chi_n(\mu) \right\|_\infty = \mu \left(\sum_{n=1}^{N} a_n e_n \right) \qquad (\mu \in \mathcal{P},\ a_n \in \mathbb{R}).$$

This extends the continuous realization theorem in the source paper from $\mathcal{B}$ to all pseudonorms. It yields a $\mathbf{\Sigma}_2^0$ reduction from all of $\mathcal{P}$ to $SB(X)$ for every isometrically universal separable Banach space X and every admissible topology, and therefore gives a positive answer to Question 2. The construction preserves all null relations, so it does not by itself answer Question 1. The argument appears not to be in the searched literature and should be independently checked before being treated as a published result.

1 Source problem and conclusions

Let $V = c_{00}(\mathbb{Q})$ with its canonical Hamel basis $(e_n)_{n\in\mathbb{N}}$. The space $\mathcal{P}$ consists of all pseudonorms on V, with the pointwise topology inherited from $\mathbb{R}^V$. Every $\mu \in \mathcal{P}$ has a unique extension to a real pseudonorm on $c_{00}(\mathbb{R})$. Let X_μ be the completion of $c_{00}(\mathbb{R})/\ker\mu$. The subspace $\mathcal{P}_\infty$ consists of codes with X_μ infinite-dimensional, and $\mathcal{B} \subseteq \mathcal{P}_\infty$ consists of those μ whose real extension is a norm. Equivalently, the vectors $[e_n]_\mu$ are linearly independent over $\mathbb{R}$ in X_μ.

The source paper asks:

Q1. Is there a continuous map $\Phi : \mathcal{P}_\infty \to \mathcal{B}$ such that $X_{\Phi(\mu)} \equiv X_\mu$ for all μ?

Q2. Given an isometrically universal separable Banach space X and an admissible topology τ on $SB(X)$, is there a $\mathbf{\Sigma}_2^0$ map $\Phi : \mathcal{P}_\infty \to (SB(X), \tau)$ such that $\Phi(\mu) \equiv X_\mu$ for all μ?

The conclusions of this audit are as follows.

1. The submitted fixed-representative statement is false as written, because rational independence does not imply real independence.

2. The same proof is correct after replacing rational independence by real independence.

3. No resolution of Question 1 was located in the literature search described in section 3; Question 1 also remains unresolved by this note.

4. Theorems 5.2 and 6.1 give a positive answer to Question 2, in the stronger form of a $\boldsymbol{\Sigma}^0_2$ reduction defined on all of $\mathcal{P}$.

2 Audit of the fixed-representative argument

Fix a strictly increasing sequence $s = (s_j)_{j\ge1}$ in $\mathbb{N}$. The proposed argument assumed that

$$\mu\left(\sum_{j=1}^{m} q_j e_{s_j}\right) > 0 \quad \text{for every nonzero } (q_1, \ldots, q_m) \in \mathbb{Q}^m, \tag{2.1}$$

and inferred that the pullback along $e_j \mapsto e_{s_j}$ belongs to $\mathcal{B}$. That inference is invalid.

Proposition 2.1 (Irrational-slope counterexample)**.** *There is $\mu \in \mathcal{P}_\infty$ such that (2.1) holds for $s_j = j$, the coordinate vectors have dense span in X_μ, but $\mu \notin \mathcal{B}$. Consequently the fixed-representative theorem in the submitted packet is false as stated.*

Proof. Fix an irrational number α, for instance $\alpha = \sqrt{2}$, and define on real c_{00}

$$\mu(x) = |x_1 - \alpha x_2| + \sum_{n=3}^{\infty} |x_n|. \tag{2.2}$$

Restricting (2.2) to V gives an element of $\mathcal{P}$. If $x \in V \setminus \{0\}$ and $\mu(x) = 0$, then $x_n = 0$ for $n \ge 3$ and $x_1 = \alpha x_2$. Since $x_1, x_2 \in \mathbb{Q}$ and $\alpha \notin \mathbb{Q}$, this forces $x_1 = x_2 = 0$, a contradiction. Thus the restriction is positive on every nonzero rational vector.

On real c_{00}, however,

$$\mu(\alpha e_1 + e_2) = 0.$$

Hence the real extension is not a norm and $\mu \notin \mathcal{B}$. The quotient map

$$c_{00} \longrightarrow c_{00}, \qquad x \longmapsto (x_1 - \alpha x_2, x_3, x_4, \ldots)$$

identifies X_μ isometrically with ℓ_1, so $\mu \in \mathcal{P}_\infty$. Finally, the images of the coordinate vectors span the canonical dense copy of $c_{00}/\ker\mu$ in its completion. □

The correct hypothesis is real linear independence. It is useful to record it in a quantitative finite-dimensional form. Put $\mathcal{P}^{\mathrm{ind}}_s$ equal to the set of all $\mu \in \mathcal{P}_\infty$ such that

$$\inf_{\|a\|_2=1} \mu\left(\sum_{j=1}^{m} a_j e_{s_j}\right) > 0 \quad \text{for every } m \in \mathbb{N}, \tag{2.3}$$

$$\overline{\operatorname{span}}\{[e_{s_j}]_\mu : j \in \mathbb{N}\} = X_\mu. \tag{2.4}$$

Condition (2.3) is equivalent to real linear independence of every finite initial segment. The strict lower bound follows from compactness of the Euclidean unit sphere.

Theorem 2.2 (Corrected fixed-stratum theorem)**.** *For every strictly increasing $s = (s_j)$, the formula*

$$\Phi_s(\mu)\left(\sum_{j=1}^{m} q_j e_j\right) = \mu\left(\sum_{j=1}^{m} q_j e_{s_j}\right), \qquad q_j \in \mathbb{Q}, \tag{2.5}$$

defines a continuous map $\Phi_s : \mathcal{P}_s^{\mathrm{ind}} \to \mathcal{B}$. Moreover, $X_{\Phi_s(\mu)} \equiv X_\mu$ for every $\mu \in \mathcal{P}_s^{\mathrm{ind}}$.

Proof. Let $T_s : c_{00} \to c_{00}$ be the real-linear map $T_s e_j = e_{s_j}$. The real extension of (2.5) is

$$\Phi_s(\mu)(x) = \mu(T_s x).$$

By (2.3), $T_s x$ is not μ-null when $x \neq 0$; hence $\Phi_s(\mu)$ is a norm and lies in $\mathcal{B}$.

For $v \in V$, the coordinate function $\mu \mapsto \Phi_s(\mu)(v) = \mu(T_s v)$ is continuous in the pointwise topology of $\mathcal{P}$. This proves continuity of Φ_s.

The map $e_j \mapsto [e_{s_j}]_\mu$ is an isometry from $(c_{00}, \Phi_s(\mu))$ into X_μ. Its range has dense span by (2.4), so it extends to a surjective linear isometry $X_{\Phi_s(\mu)} \to X_\mu$. $\square$

Corollary 2.3. *Let $A \subseteq \mathbb{N}$ have infinite complement $S = \{s_1 < s_2 < \cdots\}$. On the stratum where $[e_i]_\mu = 0$ for $i \in A$ and $([e_{s_j}]_\mu)_j$ is real-linearly independent, Question 1 has a positive answer via Φ_s.*

Proof. Every finitely supported vector is, modulo the null coordinates indexed by A, a finite linear combination of the e_{s_j}. Thus (2.4) holds, and theorem 2.2 applies. $\square$

3 Literature check

The published source paper [1] proves a $\mathbf{\Sigma}_2^0$ reduction $\mathcal{P}_\infty \to \mathcal{B}$, a continuous realization theorem on $\mathcal{B}$, and a $\mathbf{\Sigma}_3^0$ reduction $\mathcal{P} \to SB(X)$. It explicitly leaves open both Questions 1 and 2, and also states that it is open whether its continuous realization theorem extends from $\mathcal{B}$ to all of $\mathcal{P}$.

The search for a later resolution used the exact wording of both questions, combinations of the symbols $\mathcal{P}_\infty$, $\mathcal{B}$, $SB(X)$ and "admissible topology", the arXiv record and versions of the source paper, the 2024 sequel [2], the citation trail visible through the publisher and general scholarly search, and a 2026 quotient-encoding preprint that cites the source [4]. The cited papers visible in that trail through June 2026 do not advertise a solution of these questions. A 2025 thesis surveying the coding comparison [3] still records the $\mathbf{\Sigma}_3^0$ map from $\mathcal{P}$ to $SB(X)$ as the available general result. No exact published or arXiv answer was located.

This is necessarily an absence-of-evidence statement rather than a proof of novelty. The construction below should therefore be viewed as a candidate new solution of Question 2 pending expert verification.

4 A continuously varying compact dual ball

The main idea is to place the dual balls of all X_μ inside one compact Hilbert cube, prove Hausdorff continuity, and use Hilbert-space metric projection to pull the varying canonical function spaces back to one fixed $C(Q)$.

Fix positive numbers $w_n = 2^{-n}$ and set

$$Q = \{z = (z_n) \in \ell_2 : |z_n| \leq w_n \text{ for all } n\}. \tag{4.1}$$

Since $(w_n) \in \ell_2$, Q is compact in the norm topology of ℓ_2. For $\mu \in \mathcal{P}$ put

$$d_n(\mu) = 1 + \mu(e_n), \qquad u_n^\mu = \frac{[e_n]_\mu}{d_n(\mu)} \in X_\mu. \tag{4.2}$$

Then $\|u_n^\mu\| < 1$. For $N \in \mathbb{N}$ and $a = (a_1, \ldots, a_N) \in \mathbb{R}^N$, define

$$p_{\mu,N}(a) = \left\| \sum_{n=1}^N a_n u_n^\mu \right\| = \mu\left(\sum_{n=1}^N \frac{a_n}{d_n(\mu)} e_n \right). \tag{4.3}$$

Lemma 4.1 (Continuity of real finite-dimensional evaluations)**.** *For every N, the map*

$$\mathcal{P} \times \mathbb{R}^N \longrightarrow \mathbb{R}, \qquad (\mu, a) \longmapsto \mu\left(\sum_{n=1}^N a_n e_n \right)$$

is continuous. Consequently, if $\mu_j \to \mu$ in $\mathcal{P}$, then $p_{\mu_j,N} \to p_{\mu,N}$ uniformly on every compact subset of $\mathbb{R}^N$.

Proof. It is enough to use sequences. Suppose $\mu_j \to \mu$ and $a^{(j)} \to a$ in $\mathbb{R}^N$. The numbers $\mu_j(e_n)$, $j \in \mathbb{N}$ and $n \le N$, are uniformly bounded. Hence there is $M < \infty$ such that

$$\mu_j\left(\sum_{n=1}^N b_n e_n \right) \le M\|b\|_1 \quad (j \in \mathbb{N},\ b \in \mathbb{R}^N). \tag{4.4}$$

The same estimate holds for μ after increasing M if needed. Thus

$$|\mu_j(a^{(j)}) - \mu_j(a)| \le M\|a^{(j)} - a\|_1 \longrightarrow 0.$$

To prove $\mu_j(a) \to \mu(a)$, approximate a by a rational vector q and use

$$|\mu_j(a) - \mu(a)| \le \mu_j(a - q) + |\mu_j(q) - \mu(q)| + \mu(a - q),$$

where the middle term tends to zero by the defining topology of $\mathcal{P}$ and the other two terms are bounded by (4.4). This proves joint continuity. The final assertion follows by applying uniform continuity on the compact product of a convergent sequence (together with its limit) and the given compact subset of $\mathbb{R}^N$, also taking into account the continuous denominators in (4.3). $\square$

For $\mu \in \mathcal{P}$ and $N \in \mathbb{N}$, let

$$D_{\mu,N} = \{t \in \mathbb{R}^N : |\langle a, t\rangle| \le p_{\mu,N}(a) \text{ for every } a \in \mathbb{R}^N\}. \tag{4.5}$$

Lemma 4.2 (Finite polar balls)**.** *Each $D_{\mu,N}$ is a nonempty compact convex symmetric subset of $[-1,1]^N$. Its support function is*

$$h_{D_{\mu,N}}(a) = \sup_{t \in D_{\mu,N}} \langle a, t\rangle = p_{\mu,N}(a). \tag{4.6}$$

Moreover, $\mu \mapsto D_{\mu,N}$ is continuous for the Euclidean Hausdorff metric.

Proof. All structural properties follow immediately from (4.5); the coordinate bound follows by testing $a = e_n$ and observing that $p_{\mu,N}(e_n) = \mu(e_n)/(1+\mu(e_n)) \le 1$.

The inequality $h_{D_{\mu,N}} \le p_{\mu,N}$ is part of the definition. For the reverse inequality, fix $a \in \mathbb{R}^N$. Hahn–Banach, applied to the seminorm $p_{\mu,N}$, gives a linear functional ℓ on $\mathbb{R}^N$ with $|\ell(b)| \le p_{\mu,N}(b)$ for all b and $\ell(a) = p_{\mu,N}(a)$. Writing $\ell(b) = \langle b, t\rangle$ gives $t \in D_{\mu,N}$ and proves (4.6).

For compact convex subsets of Euclidean space,

$$d_{\mathrm{H}}(A,B) = \sup_{\|a\|_2 \le 1} |h_A(a) - h_B(a)|. \tag{4.7}$$

By theorem 4.1, the right-hand side for $A = D_{\mu_j,N}$ and $B = D_{\mu,N}$ tends to zero whenever $\mu_j \to \mu$. □

Define the compact convex set

$$K_\mu = \left\{ z \in Q : \left| \sum_{n=1}^{N} a_n \frac{z_n}{w_n} \right| \le p_{\mu,N}(a) \text{ for every } N \in \mathbb{N} \text{ and } a \in \mathbb{R}^N \right\}. \tag{4.8}$$

It is nonempty because $0 \in K_\mu$.

Lemma 4.3 (Dual-ball representation)**.** *Let $B_{X_\mu^*}$ be the closed dual unit ball. Then*

$$K_\mu = \left\{ \left(w_n f(u_n^\mu)\right)_{n\ge 1} : f \in B_{X_\mu^*} \right\}. \tag{4.9}$$

If $\pi_N : \ell_2 \to \mathbb{R}^N$ denotes the first-coordinate projection and $W_N = \operatorname{diag}(w_1, \dots, w_N)$, then

$$\pi_N(K_\mu) = W_N D_{\mu,N}. \tag{4.10}$$

Proof. If $f \in B_{X_\mu^*}$ and $z_n = w_n f(u_n^\mu)$, then $z \in Q$ and every inequality in (4.8) follows from

$$\left| f\left(\sum_{n=1}^{N} a_n u_n^\mu \right) \right| \le p_{\mu,N}(a).$$

Conversely, if $z \in K_\mu$, define on the algebraic span of the u_n^μ

$$f_z\left(\sum_{n=1}^{N} a_n u_n^\mu \right) = \sum_{n=1}^{N} a_n \frac{z_n}{w_n}. \tag{4.11}$$

The inequalities in (4.8) show both that (4.11) is well-defined and that $\|f_z\| \le 1$. Since the u_n^μ have dense linear span in X_μ, f_z extends continuously to an element of $B_{X_\mu^*}$. This proves (4.9).

The inclusion $\pi_N(K_\mu) \subseteq W_N D_{\mu,N}$ is immediate. Conversely, $t \in D_{\mu,N}$ defines a norm-one functional on $\operatorname{span}\{u_1^\mu, \dots, u_N^\mu\}$ by $\sum a_n u_n^\mu \mapsto \sum a_n t_n$. Hahn–Banach extends it to X_μ, and (4.9) supplies an element of K_μ whose first N coordinates are $W_N t$. □

Proposition 4.4 (Hausdorff continuity of the dual balls)**.** *The map $\mu \mapsto K_\mu$ from $\mathcal{P}$ into the nonempty compact subsets of Q is continuous for the Hausdorff metric inherited from ℓ_2.*

Proof. Set

$$r_N = \left(\sum_{n>N} w_n^2 \right)^{1/2} \longrightarrow 0.$$

For any $\mu, \nu \in \mathcal{P}$, (4.10) gives

$$d_{\mathrm{H}}(K_\mu, K_\nu) \le d_{\mathrm{H}}(W_N D_{\mu,N}, W_N D_{\nu,N}) + 2r_N. \tag{4.12}$$

Indeed, given $z \in K_\mu$, choose $y \in K_\nu$ whose first N coordinates are within the finite-dimensional Hausdorff distance of $\pi_N z$. Both tails have ℓ_2 norm at most r_N, yielding (4.12); the reverse direction is identical.

If $\mu_j \to \mu$, then for every fixed N the first term on the right of (4.12) tends to zero by theorem 4.2. Taking $\limsup_j$ and then $N \to \infty$ proves the result. $\square$

5 Metric projection and the continuous realization theorem

For a nonempty closed convex subset K of ℓ_2, let P_K denote the Hilbert-space metric projection onto K.

Lemma 5.1 (Continuous dependence of metric projections)**.** *If K_j, K are nonempty compact convex subsets of Q and $d_{\mathrm{H}}(K_j, K) \to 0$, then*

$$\sup_{z \in Q} \| \mathrm{P}_{K_j} z - \mathrm{P}_K z \|_2 \longrightarrow 0. \tag{5.1}$$

Proof. First suppose $z_j \to z$ in Q and put $p_j = \mathrm{P}_{K_j} z_j$. Compactness of Q gives convergent subsequences of (p_j). If $p_{j_k} \to p$, Hausdorff convergence implies $p \in K$. For every $y \in K$, choose $y_{j_k} \in K_{j_k}$ with $y_{j_k} \to y$. The minimizing property gives

$$\|z_{j_k} - p_{j_k}\|_2 \le \|z_{j_k} - y_{j_k}\|_2.$$

Passing to the limit shows $\|z - p\|_2 \le \|z - y\|_2$ for all $y \in K$. Uniqueness of Hilbert-space projection yields $p = \mathrm{P}_K z$. Thus $\mathrm{P}_{K_j} z_j \to \mathrm{P}_K z$.

If (5.1) failed, there would be $z_j \in Q$ and $\varepsilon > 0$ with $\| \mathrm{P}_{K_j} z_j - \mathrm{P}_K z_j\|_2 \ge \varepsilon$. Passing to a subsequence, compactness of Q gives $z_j \to z$, and the preceding sequential continuity applied both to K_j and to the constant sequence K gives a contradiction. $\square$

For $\mu \in \mathcal{P}$, abbreviate

$$P_\mu = \mathrm{P}_{K_\mu}|_Q : Q \longrightarrow K_\mu.$$

Since P_μ fixes every point of K_μ, it maps Q onto K_μ.

Theorem 5.2 (Continuous realization of all pseudonorms)**.** *There is a fixed compact metric space Q and continuous maps*

$$\chi_n : \mathcal{P} \longrightarrow C(Q), \qquad n \in \mathbb{N},$$

such that for every $\mu \in \mathcal{P}$, every $N \in \mathbb{N}$ and every $a_1, \dots, a_N \in \mathbb{R}$,

$$\left\| \sum_{n=1}^{N} a_n \chi_n(\mu) \right\|_{C(Q)} = \mu\left(\sum_{n=1}^{N} a_n e_n \right). \tag{5.2}$$

One may take Q as in (4.1) and

$$\chi_n(\mu)(z) = d_n(\mu) \frac{(P_\mu z)_n}{w_n}, \qquad z \in Q. \tag{5.3}$$

Proof. For fixed μ and n, formula (5.3) defines a continuous function on Q. If $\mu_j \to \mu$, then theorems 4.4 and 5.1 give $P_{\mu_j} \to P_\mu$ uniformly on Q, while $d_n(\mu_j) \to d_n(\mu)$. Hence $\chi_n(\mu_j) \to \chi_n(\mu)$ in the supremum norm, proving continuity of χ_n.

Because $P_\mu(Q) = K_\mu$, theorem 4.3 gives

$$\begin{aligned}\left\|\sum_{n=1}^{N} a_n\chi_n(\mu)\right\|_\infty &= \sup_{y\in K_\mu}\left|\sum_{n=1}^{N} a_n d_n(\mu)\frac{y_n}{w_n}\right| \\ &= \sup_{f\in B_{X_\mu^*}}\left|f\left(\sum_{n=1}^{N} a_n[e_n]_\mu\right)\right| \\ &= \left\|\sum_{n=1}^{N} a_n[e_n]_\mu\right\|_{X_\mu} = \mu\left(\sum_{n=1}^{N} a_n e_n\right),\end{aligned}$$

where the penultimate equality is the usual dual formula for the norm. This is (5.2). ☐

Remark 5.3. *The source paper proves the analogue of theorem 5.2 on $\mathcal{B}$ and explicitly asks whether it extends to $\mathcal{P}$. Thus theorem 5.2 settles that intermediate problem as well. Conceptually, K_μ is a weighted copy of the weak-star compact dual unit ball, and P_μ is a continuously varying surjection from one fixed compact space onto K_μ.*

6 Positive answer to Question 2

Let X be an isometrically universal separable Banach space. Since $C(Q)$ is separable, fix a linear isometric embedding $J : C(Q) \to X$. Define

$$F(\mu) = \overline{\operatorname{span}}\{J\chi_n(\mu) : n \in \mathbb{N}\} \in SB(X). \tag{6.1}$$

Corollary 6.1 (Question 2, with a stronger domain)**.** *For every admissible topology τ on $SB(X)$, the map*

$$F : \mathcal{P} \longrightarrow (SB(X), \tau)$$

is $\mathbf{\Sigma}^0_2$-measurable and satisfies $F(\mu) \equiv X_\mu$ for every $\mu \in \mathcal{P}$. Its restriction to $\mathcal{P}_\infty$ takes values in $SB_\infty(X)$ and gives a positive answer to Question 2.

Proof. By (5.2), the linear map $e_n \mapsto J\chi_n(\mu)$ has exactly the pseudonorm μ. It therefore factors through $c_{00}/\ker\mu$ and extends to a surjective linear isometry $X_\mu \to F(\mu)$.

For an open set $U \subseteq X$, write $E^+(U) = \{Y \in SB(X) : Y \cap U \neq \varnothing\}$. Rational finite linear combinations of the $J\chi_n(\mu)$ are dense in $F(\mu)$, so

$$F^{-1}(E^+(U)) = \bigcup_{q\in V}\left\{\mu \in \mathcal{P} : \sum_n q_n J\chi_n(\mu) \in U\right\}. \tag{6.2}$$

Every set on the right is open, hence the inverse image in (6.2) is open.

By definition, an admissible topology has a subbasis whose members are countable unions of sets $E^+(U) \setminus E^+(V)$ with U, V open in X. The inverse image of such a difference is the difference of two open subsets of the Polish space $\mathcal{P}$. This is an F_σ set: every open subset of a metrizable space is F_σ, and intersecting an F_σ set with a closed set preserves that class. Therefore inverse images of subbasic open sets are $\mathbf{\Sigma}^0_2$. Since the target topology is second countable, every open set is a countable union of finite intersections of subbasic sets, and $\mathbf{\Sigma}^0_2$ is closed under these operations. Thus F is $\mathbf{\Sigma}^0_2$-measurable. ☐

Remark 6.2. *The conclusion is stronger than Question 2 in two ways: the map is defined on all pseudonorms, including finite-dimensional and zero codes, and the hit-open sets in* (6.2) *have open inverse image. The loss to $\mathbf{\Sigma}^0_2$ occurs only when passing from the lower hit topology to an arbitrary admissible topology.*

7 What remains for Question 1

The realization theorem deliberately preserves null relations. Indeed, (5.2) gives

$$\sum_{n=1}^{N} a_n\chi_n(\mu) = 0 \quad \iff \quad \mu\left(\sum_{n=1}^{N} a_n e_n\right) = 0. \tag{7.1}$$

Thus, when $\mu \notin \mathcal{B}$, the displayed continuous generating sequence is not linearly independent. To answer Question 1 one must additionally replace it, continuously in μ, by a real-linearly independent dense sequence in the same Banach space. Neither the metric-projection construction nor the corrected fixed-stratum theorem performs that global selection.

The following observation rules out the most rigid possible repair.

Proposition 7.1 (No fixed real-linear pullback solves Question 1)**.** *There is no fixed real-linear map $T : c_{00}(\mathbb{R}) \to c_{00}(\mathbb{R})$ such that*

$$\Phi_T(\mu)(v) = \mu(Tv), \qquad v \in V,$$

belongs to $\mathcal{B}$ and codes a space isometric to X_μ for every $\mu \in \mathcal{P}_\infty$.

Proof. If T is not injective, choose $0 \neq v \in c_{00}(\mathbb{R})$ with $Tv = 0$; then the real extension of $\Phi_T(\mu)$ has the nonzero null vector v for every μ. Suppose T is injective and choose $0 \neq v \in c_{00}(\mathbb{R})$. Put $y = Tv \neq 0$. Choose a real-linear automorphism A of c_{00} with $Ay = e_1$, and define

$$\mu(x) = \sum_{n=2}^{\infty} |(Ax)_n|.$$

Then $\ker \mu = \mathbb{R}y$, while X_μ is infinite-dimensional (indeed, isometric to ℓ_1 after a coordinate identification). Hence $\Phi_T(\mu)(v) = \mu(y) = 0$, so $\Phi_T(\mu) \notin \mathcal{B}$. $\square$

A recursive construction would require continuous choices outside varying finite-dimensional spans. This resembles a zero-avoiding continuous-selection problem. General selection theory does not provide such avoidance on arbitrary infinite-dimensional parameter spaces without additional hypotheses. This is consistent with, but of course does not prove, the continued difficulty of Question 1.

8 Final status

The fixed-representative packet contains a correct idea but an incorrect independence test. After replacing rational independence by real independence, its theorem is valid on the stated fixed strata. The global norm-code problem (Question 1) remains open in this investigation.

The compact-dual-ball and metric-projection construction gives a candidate continuous realization of every pseudonorm in one fixed separable Banach space. Subject to independent verification, it supplies a full positive solution of Question 2 and improves the previously stated $\mathbf{\Sigma}^0_3$ reduction from $\mathcal{P}$ to $SB(X)$ to $\mathbf{\Sigma}^0_2$. Because this appears to be a new argument, independent expert checking is especially important; the most delicate points are the Hausdorff continuity in theorem 4.4 and the uniform projection continuity in theorem 5.1.

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